

A7 · Answer Key and Demo Script

CS425 · One Hop at a Time · instructor led

INSTRUCTOR COPY

The round 1 and 2 transcripts are real output captured on `onyx.boisestate.edu` on October 4, 2026. Rounds 3 and 4 are paper and were checked by script. About 40 minutes: 8, 10, 10, 10, then 2 for the exit question.

Say this at the start

"In A4 Onyx picked the next hop. Today is how the datagram actually gets across that one hop. Every frame carries two pairs of addresses, and by the end you will know which pair changes at every router. Everyone `ssh onyx`. Predict before we run anything." Then hand out the sheet.

ROUND 1 · ONYX'S OTHER ADDRESS

ip -br link		
lo	UNKNOWN	00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
eno8303	DOWN	e8:cf:83:2c:2c:b9 <NO-CARRIER,BROADCAST,MULTICAST,UP>
eno8403	DOWN	e8:cf:83:2c:2c:ba <NO-CARRIER,BROADCAST,MULTICAST,UP>
eno12399np0	UP	6c:92:cf:db:68:50 <BROADCAST,MULTICAST,UP,LOWER_UP>
eno12409np1	DOWN	6c:92:cf:db:68:51 <NO-CARRIER,BROADCAST,MULTICAST,UP>
SAY "The IP address says where Onyx is. The MAC address says who Onyx is, wherever it goes."		

Predict. Only the IP address changes. The MAC is flat and belongs to the card, not the location. That is the postal address versus social security number analogy from the notes.

OUI and bits. 6c:92:cf is Broadcom. The first byte 0x6c = 0110 1100: lowest bit 0 (unicast), next bit 0 (globally unique).

FAST GROUP The two DOWN ports start e8:cf:83, which is Dell: the ports built into the server. The Broadcom ports are an add-in card. Each card's two ports have consecutive addresses (...50, ...51), because a vendor hands out a block per card.

Phone. A randomized address has the second lowest bit of the first byte set (the first byte's hex digit pair ends in 2, 6, A or E), so it is locally administered. **Q1.** Privacy. A fixed hardware address lets every Wi-Fi network, store and stadium recognize the same phone wherever it goes. A different made up address per network breaks that tracking.

ROUND 2 · ARP

ip neigh show dev eno12399np0 then ping -c 1 8.8.8.8, ip neigh show 8.8.8.8, ip neigh show 132.178.227.1	
132.178.227.1 lladdr 0c:d0:f8:cf:ec:eb REACHABLE	
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data. --- 8.8.8.8 ping statistics --- 1 packets transmitted, 0 received, 100% packet loss, time 0ms	
132.178.227.1 dev eno12399np0 lladdr 0c:d0:f8:cf:ec:eb REACHABLE	
POINT AT the empty output for 8.8.8.8, then at the gateway's line. The ping went to the gateway's MAC. The 100% loss is something past the gateway dropping ICMP. It does not change the frame that left Onyx.	

FIELD	ADDRESS	CHANGES AT THE NEXT HOP?
Source MAC	6c:92:cf:db:68:50	Yes: becomes the gateway's outgoing interface
Destination MAC	0c:d0:f8:cf:ec:eb	Yes: becomes the next router's
Source IP	132.178.227.11	No (unless a NAT rewrites it)
Destination IP	8.8.8.8	No

Q2. ARP only works on the local link, and 8.8.8.8 is not on Onyx's /25 (A4 round 2: it matches the default route, via 132.178.227.1). So Onyx ARPs for the gateway instead, and the frame carries the gateway's MAC with Google's IP inside. Point at P4: these are the four fields they will pull out of real frames.

GRADING

Pass/fail, 20 points. Pass when every box has something in it and the round 3 division was attempted. Wrong predictions are fine; that is what they are for.

ROUND 3 · CRC BY HAND

11010011 000 divided by 1011, quotient 11110001, each line is the dividend after one XOR	
1011) 11010011000 1011 01100011000 1011 00111011000 1011 00010111000 1011 00000001000 1011 0000000011 R = 011	
SAY "Line the generator up under the leftmost 1, XOR, repeat. When there is no 1 left in the first eight columns, the last three bits are R."	

Predict. $r = 3$ bits, one less than the 4 bit generator.

R. 011. On the wire: 11010011 011. The receiver divides 11010011011 by 1011 and gets 000: no error.

Flip the third bit. 11110011011 leaves remainder 010, not zero, so the receiver drops the frame. Every single bit error is caught, since the generator has more than one term.

FAST GROUP Find an error the CRC misses. Any error pattern that is a multiple of G gets through: flip bits 2, 4 and 5 (pattern 01011000000, which is G shifted) and the remainder is 000.

ROUND 4 · BE THE SWITCH

#	FRAME	OUT WHICH PORTS?	TABLE AFTER
1	A to D	2, 3, 4 (D unknown: flood)	A:1
2	D to A	1	A:1, D:4
3	B to D	4	A:1, D:4, B:2
4	C to broadcast	1, 2, 4 (broadcast always floods)	A:1, D:4, B:2, C:3
5	A to C	3	unchanged

Flooded. 2 of 5: frame 1 (unknown destination) and frame 4 (broadcast). The switch learns from the **source** address, never the destination, which is why frame 1 teaches it about A and not D.

Q3. B sees frames 1 and 4 (flooded) plus frame 3, which it sent. It never sees 2 or 5. Behind a hub B would see all five, because a hub repeats every bit out every port. That is why Wireshark on a switched network mostly shows your own traffic and broadcasts.

EXIT QUESTION

ARP An ARP request is a link layer broadcast, not an IP datagram, so a router has nothing to forward it on and it stops at the edge of the subnet. DHCP starts as a broadcast too, but on campus a DHCP relay on the router usually forwards it to a central server, so accept DHCP only if they explain why. DNS, TCP and HTTP are ordinary datagrams and cross routers all day.